Influenza Outbreak Detection Using Machine Learning

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1 Introduction

Influenza outbreaks are a major challenge for public health systems because of the seasonal nature and effects which rapidly increase healthcare demand. Early detection of increases in influenza activity allows health authorities to use resources and implement interventions more effectively. Machine learning methods have increasingly been used for detecting and forecasting infectious disease outbreaks using surveillance data (Babanejaddehaki, An, & Papagelis, 2025).

The objective of this study was to make a machine learning model to find periods of increased influenza activity using surveillance indicators. This is a binary classification problem where observations were labelled as either baseline influenza activity or outbreak activity. Two models were used and compared: Logistic Regression and Random Forest. Detecting outbreak conditions using surveillance data can support early public health responses and improve resource allocation during periods of increased influenza activity.

2 Dataset and Preprocessing

The dataset contained influenza surveillance indicators including laboratory-confirmed influenza case counts and testing activity. Each observation represents influenza activity recorded during a specific time period.

Initial preprocessing included examining missing values and removing variables with alot of missingness. Handling incomplete variables is important in healthcare datasets because poor data quality can reduce model reliability and prediction accuracy (Salmi, Atif, Oliva, Abraham, & Ventura, 2024).

The primary variables examined included:

- INF_A: number of confirmed Influenza A cases
- INF_B: number of confirmed Influenza B cases
- SPEC_PROCESSED_NB: number of processed laboratory specimens
- ISO_WEEK: week number representing seasonality

These indicators are typical surveillance signals used to monitor influenza activity and detect emerging outbreaks (Wei et al., 2023).

3 Exploratory Data Analysis

Exploratory data analysis was required to understand the distributions and relationships between the variables. Histograms showed that influenza case counts were highly right-skewed with most observations representing low levels of activity and occasional spikes corresponding to outbreak periods. This pattern is typical for epidemiological surveillance data where most time periods represent baseline activity and occasional spikes correspond to outbreak events.

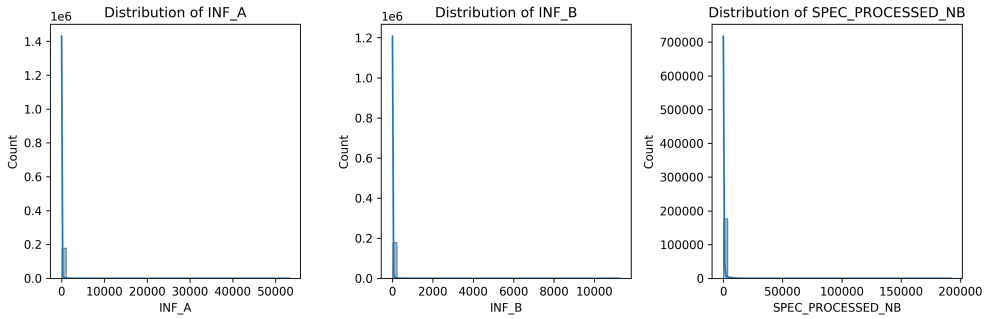

Figure 1: Distribution of influenza indicators.

A correlation matrix was created to examine relationships between variables.

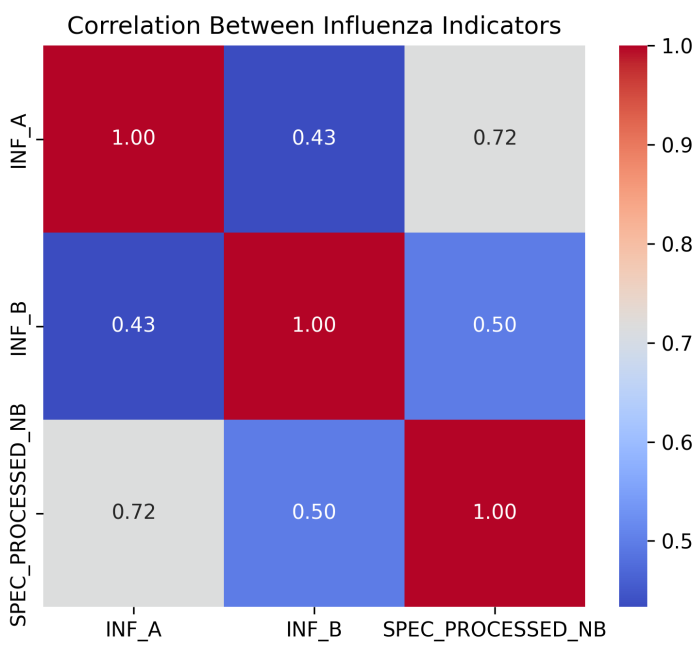

Figure 2: Correlation between influenza indicators.

4 Outbreak Definition

To train a supervised model a binary outbreak label was created. Total influenza activity was calculated as the sum of Influenza A and Influenza B cases.

$$TOTAL_INF = INF_A + INF_B \tag{1}$$

A threshold based on the 75th percentile of total influenza activity was used to define outbreak periods. Observations above this threshold were labelled as outbreaks. Percentile-based thresholds are data-driven methods for identifying unusually high activity levels in epidemiological surveillance data (Martin-Moreno et al., 2022).

This definition is an indicator of unusually high influenza activity rather than a clinically confirmed outbreak definition.

5 Feature Selection

Feature selection was based on epidemiological reasoning and machine learning considerations. Influenza A and Influenza B counts were excluded from the predictor variables because they were used to make the outbreak label. Including them as predictors would introduce data leakage and artificially inflate model performance.

The variable total influenza activity was also excluded for the same reason because it was used to make the target variable.

Operational indicators such as the number of processed laboratory specimens were used as predictors. Testing activity can reflect changes in disease transmission patterns and is often used as a proxy signal in surveillance systems (Wei et al., 2023).

A time-based feature representing the week of the year was also included to capture seasonal influenza patterns.

Because week numbers are cyclical rather than linear, the week variable was encoded using sine and cosine transformations so that the beginning and end of the year are represented correctly.

6 Machine Learning Models

Two machine learning models were used.

6.1 Logistic Regression

Logistic Regression was used as a baseline classification model. Logistic Regression is widely used in healthcare prediction because of its interpretability and how it can model the probability of binary outcomes (Wu et al., 2025).

6.2 Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees to improve predictive performance. Ensemble models are useful for capturing complex nonlinear relationships between predictors and the target variable which leads to improved prediction accuracy compared to traditional statistical models (Wu et al., 2025).

7 Hyperparameter Tuning

Hyperparameter tuning was carried out using GridSearchCV to optimize the Random Forest model. Paramaters like the number of trees and the maximum tree depth were used.

Recall was used as the optimisation metric during model tuning because outbreak detection involves identifying rare events. Failing to detect an outbreak may be more harmful than generating false positive alerts in surveillance systems.

The best performing configuration used 200 trees, a maximum tree depth of 10 and balanced class weights.

8 Model Evaluation

Model performance was evaluated using multiple metrics including accuracy, precision, recall, and F1-score. Using several evaluation metrics provides a more comprehensive understanding of classification performance in datasets where class imbalance may exist (Salmi et al., 2024).

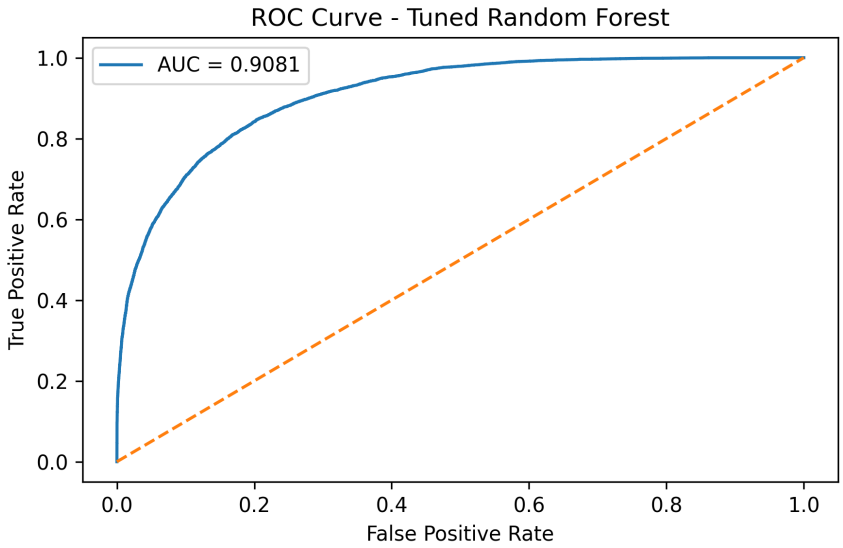

Figure 3: ROC curve for the tuned Random Forest model.

The ROC curve shows the relationship between the true positive rate and the false positive rate and helps evaluate how well the model separates outbreak periods from baseline activity.

9 Final Model Selection

The tuned Random Forest model achieved the best overall performance. Compared with Logistic Regression the Random Forest model showed improved ability to capture complex relationships between influenza indicators and outbreak conditions. Previous studies have also shown that machine learning models can outperform traditional regression models when modelling complex healthcare data. The precision–recall curve further shows the models ability to keep relatively high precision while detecting a large proportion of outbreak cases. (Wu et al., 2025).

10 Conclusion

This study showed the use of machine learning techniques to analyse influenza surveillance data and detect potential outbreak periods. Logistic Regression and Random Forest models were used and evaluated using multiple classification metrics.

The tuned Random Forest model achieved the best performance and was selected as the final model. The results highlight the potential of machine learning methods to support public health surveillance and assist in the early detection of infectious disease outbreaks.

References

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